Calculus 3

Exam 3 | Line integrals

1 Curl and divergence

36 Surface parametrizations

37–41 Concepts and method choice

LINE INTEGRALS $\int_C$

Scalar field f

Direct parametrization

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,\|\mathbf{r}'(t)\|\,dt.$$

Arc length: $f = 1$.

Wire mass: $f = \rho$ (linear density).

Exam 2 review

Vector field: circulation / work

Is $\mathbf{F}$ conservative?
Is there a potential ϕ with $\mathbf{F} = \nabla\phi$?

Yes

Fundamental theorem of line integrals

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(\mathbf{r}(b)) - \phi(\mathbf{r}(a)).$$

If C is closed, the integral is 0.

2–5 **41(2), (6): closed curves**

No

Is C a closed curve?
Same starting and ending point?

No

Direct parametrization

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt.$$

Works for open *or* closed curves.

Also $d\mathbf{r} = \mathbf{T}\,ds$.

Yes

Choose a simpler spanning region or surface.

Planar (2D)

Green's theorem

$\mathbf{F} = \langle P, Q\rangle$ and $C = \partial D$.

Circulation: C counterclockwise

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_D (Q_x - P_y)\,dA.$$

Also, planar flux: $\mathbf{n}$ the outward unit normal

$$\oint_C \mathbf{F}\cdot\mathbf{n}\,ds = \iint_D (P_x + Q_y)\,dA.$$

6–13

Spatial (3D)

Stokes' theorem

For $C = \partial S$ with compatible orientations,

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS.$$

Choose a spanning surface that makes the curl flux easy to compute.

Stokes' theorem also applies to planar surfaces.

41: method choice

Check the domain before using a potential

For a C^1 field on an open, simply connected domain, zero curl implies a potential. In two dimensions, use $Q_x = P_y$. Without the domain hypothesis, zero curl need not imply conservative.

Method choice: direct parametrization works for closed curves too. Closed does *not* mean zero circulation.

References. FTLI: 2–5. Green's theorem: 6–13. Conceptual method selection: 40–41. Scalar line integrals are Exam 2 review.

Question 41 convention. The listed choices refer to direct use of the formulas on the given integral. Rewriting a spatial line integral in planar form can make additional methods available.

Conditions. Use smooth fields and piecewise smooth curves; Green's circulation uses the positive boundary orientation.

Calculus 3

Exam 3 | Surface integrals

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SURFACE INTEGRALS $\iint_S$

Scalar field f

Direct parametrization

Parametrize S by $\mathbf{r}(u,v)$, $(u,v) \in D$.

$$\iint_S f\,dS = \iint_D f(\mathbf{r}(u,v)) \cdot \|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv.$$

Surface area: set $f = 1$.

Mass: use $f = \rho$ (surface density).

Orientation does not affect a scalar surface integral.

14–22

Vector field

Which flux is given?
The flux of $\mathbf{F}$, or the flux of $\nabla \times \mathbf{F}$?

$\mathbf{F}$

Surface flux of $\mathbf{F}$

$$\iint_S \mathbf{F} \cdot \mathbf{n}\,dS = \iint_D \mathbf{F}(\mathbf{r}(u,v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,du\,dv.$$

Choose the cross product to match the specified orientation. Reverse it if necessary.

This flux **can be nonzero on a closed surface**.

Coordinator's example: HW9, question 8.

23–28

$\nabla \times \mathbf{F}$

Is S closed?
$\partial S = \varnothing$ (no boundary)

Yes

Closed-surface curl flux

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = 0.$$

No boundary, hence no boundary circulation.

The field must be defined and smooth near all of S.

34–35

No

Stokes' theorem

For $C = \partial S$ with compatible orientations,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Compute the boundary circulation, or replace S by an easier spanning surface.

29–33

Same oriented boundary: change the surface, not the answer

$$\partial S = \partial \widetilde{S} = C \quad \Longrightarrow \quad \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,dS = \iint_{\widetilde{S}} (\nabla \times \mathbf{F}) \cdot \widetilde{\mathbf{n}}\,dS.$$

Example: an upward-oriented hemisphere can be replaced by its upward-oriented equatorial disk. The field must be smooth near both surfaces. This rule concerns **curl flux**, not arbitrary flux.

Notation. $\mathbf{r}(u,v)$ parametrizes S over D; $\mathbf{n}$ is the specified unit normal. In flux formulas, the oriented cross product includes the area factor.

Conditions. Use smooth fields near the surfaces and compatible boundary orientations. For practice 29, use the intended hemisphere including its rim ($z \geq 0$).

References. Scalar surfaces: 14–22. Flux: 23–28. Stokes: 29–35.

Formula sheet

1. Coordinate Conversion Formulas

Cylindrical to Rectangular	Spherical to Rectangular	Spherical to Cylindrical
$x = r\,cos(\theta)$ $y = r\,sin(\theta)$ $z = z$ $x^2 + y^2 = r^2$	$x = \rho \sin(\varphi)\,cos(\theta)$ $y = \rho \sin(\varphi)\,sin(\theta)$ $z = \rho cos(\varphi)$ $x^2 + y^2 + z^2 = \rho^2$	$r = \rho\,sin(\varphi)$ $z = \rho \cos(\varphi)$ $\theta = \theta$

2. Fundamental Theorem of Line integrals:

$$\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = f(B) - f(A)$$

where *f is* a potential function of vector field **F**, and A, B are end points of curve *C*.

3. Green's Theorem

Let $\boldsymbol{F}(x,y) = P(x,y)\,\boldsymbol{i} + Q(x,y)\,\boldsymbol{j}$ be a vector field. Then,

- $\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = \iint_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA$
- $\int_C \boldsymbol{F} \cdot \boldsymbol{N} ds = \iint_R \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\, dA$

4. Stokes' Theorem

$$\int_C \boldsymbol{F} \cdot \boldsymbol{T} ds = \iint_S (\nabla \times \boldsymbol{F}) \cdot \boldsymbol{N}\, dS$$

Exam 3 Content Overview:

- **Line integrals** include the Fundamental Theorem of Line Integrals for conservative vector fields and Green's Theorem.
- **Surface integrals** include parameterization of surfaces, surface integrals of scalar functions, and surface integrals of vector fields (i.e., flux integrals).
- **Stokes' Theorem** connects the two: it equates the line integral (circulation) of $\mathbf{F}$ around a closed curve C to the surface integral (flux) of $\operatorname{curl}\mathbf{F}$ through any oriented surface S with boundary C.

Line Integrals:

1. **Fundamental Theorem of Line Integrals (conservative vector fields)**

   If the vector field $\mathbf{F} = P(x,y,z)\,\mathbf{i} + Q(x,y,z)\,\mathbf{j} + R(x,y,z)\,\mathbf{k}$ is **conservative**, i.e., $\mathbf{F} = \nabla f$, then the line integral of $\mathbf{F}$ along **any curve** C from A to B is

   $$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A).$$

   The line integral is path-independent, and $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C.

   Procedure:

   Step 1: Verify $\mathbf{F}$ is conservative by checking $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F} = \mathbf{0}$.

   Step 2: Find a potential function f such that $\mathbf{F} = \nabla f$.

   Step 3: Plug initial and terminal points, A and B, of the curve into f.

   Practice: Q1–5

2. **Green's Theorem**

   Let R be a region in the xy-plane, and let C be its boundary, a simple **closed** curve traversed counterclockwise (positive orientation). For a **2D** vector field $\mathbf{F} = P(x,y)\,\mathbf{i} + Q(x,y)\,\mathbf{j}$:

   - ↘ Circulation (tangential) form:

     $$\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

     Compute the double integral using either rectangular or polar coordinates, whichever is easier.

   - ↘ Flux (normal) form:

     $$\oint_C \mathbf{F} \cdot \mathbf{N}\, ds = \iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA,$$

     where $\mathbf{N}$ is the outward unit normal to C.

Practice: Q6–13

Surface Integrals:

1. **Parameterization of Surfaces**

   Find parameterization equations for surfaces (sphere, cone, cylinder, plane, surface of revolution, circular disk, general surfaces).

   Practice: Q36

2. **Surface integrals of scalar functions**

   ✓ To set up $\iint_S f(x,y,z)\,d\sigma$:

   Step 1: Find a parameterization for the surface, $\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$.

   Step 2: Calculate $\mathbf{r}_u \times \mathbf{r}_v$.

   Step 3: Calculate the magnitude $|\mathbf{r}_u \times \mathbf{r}_v|$.

   Step 4:

   $$\iint_S f(x,y,z)\,d\sigma = \iint_D f\big(x(u,v), y(u,v), z(u,v)\big)\,|\mathbf{r}_u \times \mathbf{r}_v|\,du\,dv,$$

   where D is the parameter domain in the uv-plane.

   ✓ Applications: (1) Surface area: take $f = 1$, so Area $= \iint_S 1\,d\sigma$. (2) Mass of a surface with density $\rho(x,y,z)$: Mass $= \iint_S \rho\,d\sigma$.

   Practice: Q14–22

3. **Surface integrals of vector fields** (Flux Integrals)

   To compute $\iint_S \mathbf{F}(x,y,z)\cdot\mathbf{N}\,d\sigma$:

   Step 1: Find a parameterization for the surface, $\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$.

   Step 2: Calculate $\mathbf{r}_u \times \mathbf{r}_v$.

   Step 3: Check whether the normal vector $\mathbf{r}_u \times \mathbf{r}_v$ agrees with the prescribed orientation of S (e.g., upward or outward unit normal $\mathbf{N}$). If not, reverse its sign.

   Step 4:

   $$\iint_S \mathbf{F}(x,y,z)\cdot\mathbf{N}\,d\sigma = \iint_D \mathbf{F}\big(x(u,v), y(u,v), z(u,v)\big)\cdot\mathbf{N}\,|\mathbf{r}_u \times \mathbf{r}_v|\,du\,dv,$$

   where D is the parameter domain.

Step 5: Evaluate the double integral with respect to u and v.

Practice: Q23–28

Stokes' Theorem (connecting line and surface integrals)

Stokes' Theorem: For **3D** vector fields $\mathbf{F} = P(x,y,z)\,\mathbf{i} + Q(x,y,z)\,\mathbf{j} + R(x,y,z)\,\mathbf{k}$, and a piecewise-smooth, simple **closed** curve $C = \partial S$ bounding an oriented surface S, where C carries the induced (positive) orientation, consistent with $\mathbf{N}$ by the right-hand rule:

$$\oint_C \mathbf{F}\cdot\mathbf{T}\,ds = \iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$$

Practice: Q29–35.

Exam 3 focus: convert the flux integral of $\operatorname{curl}\mathbf{F}$ across S into the circulation of $\mathbf{F}$ around its boundary curve $C = \partial S$, i.e., evaluate $\iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$ by computing $\oint_C \mathbf{F}\cdot d\mathbf{r}$.

Final exam: the reverse direction will also be tested: evaluate a circulation $\oint_C \mathbf{F}\cdot d\mathbf{r}$ by computing $\iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{N}\,d\sigma$ over a suitable surface S bounded by C.

Conceptual Questions:

- Identify which compositions of gradient, curl, and divergence are well-defined, based on whether each operator acts on a scalar field or a vector field. "Select all that apply" **(Practice 37, HW9 Problem 19)**.
- The physical interpretation of curl (circulation density, i.e., local rotation) and divergence (flux density per unit volume: a source if $\operatorname{div}\mathbf{F} > 0$, a sink if $\operatorname{div}\mathbf{F} < 0$). **(Practice 38–39)**
- Choose among the Fundamental Theorem of Line Integrals, Green's Theorem, Stokes' Theorem, and the direct parameterization method: select all methods that could evaluate a given line integral. **(Practice 40–41; see the decision guide below.)**

Choosing a method for a line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$:

- $\mathbf{F}$ is **conservative** ($\operatorname{curl}\mathbf{F} = \mathbf{0}$) $\Rightarrow$ **Fundamental Theorem of Line Integrals**: $\int_C \mathbf{F}\cdot d\mathbf{r} = f(B) - f(A)$, for **any curve** C (open or closed; the integral is 0 if C is closed).
- C is **closed** and $\mathbf{F}$ is a **2D** field $\Rightarrow$ **Green's Theorem**: convert to a double integral over the enclosed region R.
- C is **closed** and $\mathbf{F}$ is a **3D** field $\Rightarrow$ **Stokes' Theorem**: convert to the flux of $\operatorname{curl}\mathbf{F}$ through a surface S with $\partial S = C$.
- **Any** $\mathbf{F}$, **any curve** C $\Rightarrow$ **Direct parameterization**: $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\mathbf{r}'(t)\,dt$. This method always works; the theorems above are shortcuts. It is the only option when $\mathbf{F}$ is not conservative and C is **not closed**.

Exam 3 Practice Problems — Answers

Equivalent mathematical forms are valid. In planar flux questions, the normal is outward.

1. (a) $\nabla \times \mathbf{F} = (8y + 3z^2, 0, 3 - x^2)$, $\nabla \cdot \mathbf{F} = 2xy$. Not conservative.

   (b) $\nabla \times \mathbf{F} = \left(\dfrac{x^3y^2}{z^2}, 4z - 7, \dfrac{3x^2y^2}{z} \right)$, $\nabla \cdot \mathbf{F} = 2 + \dfrac{2x^3y}{z}$. Not conservative on either half-space $z > 0$ or $z < 0$.

   (c) $\nabla \times \mathbf{F} = (0, 3x^2, 0)$, $\nabla \cdot \mathbf{F} = 14 - 2y$. Not conservative.

   (d) $\nabla \times \mathbf{F} = (0, 0, 0)$, $\nabla \cdot \mathbf{F} = -xz \cos y$. Conservative, with potential $f = xz \cos y + C$.

2. Conservative: $\nabla \times \mathbf{F} = \mathbf{0}$ and $\boxed{f = x^2y^2 + xz + C}$. Indeed, $P_y = Q_x = 4xy$, $P_z = R_x = 1$, and $Q_z = R_y = 0$.

3. Conservative, with $\boxed{f = xyz + e^x + e^y + e^z + C}$.

4. $\mathbf{F} = \nabla(xyz^2)$, so the integral is $f(1,1,1) - f(0,0,0) = \boxed{1}$.

5. $\boxed{3e - 2}$, using the potential from Problem 3.

6. $Q_x - P_y = -1 - 2y$, so
$$\int_0^1 \int_0^x (-1 - 2y) \, dy \, dx = \boxed{-\frac{5}{6}}.$$

7. $P_x + Q_y = 0$, so the outward flux is $\boxed{0}$.

8. $\boxed{-8\pi}$.

9. $\boxed{8\pi}$.

10. $\boxed{1}$.

11. $\boxed{8}$.

12. $Q_x - P_y = y$, so
$$\int_0^1 \int_{x^2}^x y \, dy \, dx = \frac{1}{2} \int_0^1 (x^2 - x^4) \, dx = \boxed{\frac{1}{15}}.$$

13. $P_x + Q_y = 3x$, so
$$\int_0^1 \int_{x^2}^x 3x \, dy \, dx = \boxed{\frac{1}{4}}.$$

14.

$$\mathbf{r}_u \times \mathbf{r}_v = (2u^2 + 2v^2, v - u, -2(u + v)),$$

$$\boxed{A = \int_0^2 \int_0^1 \sqrt{(2u^2 + 2v^2)^2 + (v - u)^2 + 4(u + v)^2}\ \mathrm{d}u\ \mathrm{d}v}.$$

15.

$$\mathbf{r}_u \times \mathbf{r}_v = (v^2 - 2uv, u^2 - 2uv, 3u^2v^2),$$

$$\boxed{m = \int_0^1 \int_0^1 e^{u+v}\sqrt{(v^2 - 2uv)^2 + (u^2 - 2uv)^2 + 9u^4v^4}\ \mathrm{d}u\ \mathrm{d}v}.$$

16.

$$\boxed{A = 2\pi \int_0^1 e^{5x}\sqrt{1 + 25e^{10x}}\ \mathrm{d}x}.$$

17. Interpreting the scalar function on the spatial surface as $f(x, y, z) = y^5$,

$$\boxed{\iint_S f\ \mathrm{d}\sigma = 2\pi \int_1^4 x^{-3/2}\sqrt{1 + \frac{1}{4x^3}}\ \mathrm{d}x}.$$

The radius is x, the density is $(x^{-1/2})^5$, and the generating curve has speed $\sqrt{1 + 1/(4x^3)}$.

18. $\boxed{\pi/3}$.

19. Use $\mathbf{r}(r, \theta) = (r\cos\theta, r\sin\theta, r)$, with $2 \le r \le 4$, $0 \le \theta \le 2\pi$. Then

$$\iint_S z^2\ \mathrm{d}\sigma = \int_0^{2\pi} \int_2^4 r^2\sqrt{2}\, r\ \mathrm{d}r\ \mathrm{d}\theta = \boxed{120\sqrt{2}\,\pi}.$$

20.

$$A = \int_0^{2\pi} \int_0^1 r\sqrt{1 + 4r^2}\ \mathrm{d}r\ \mathrm{d}\theta = \boxed{\frac{\pi(5\sqrt{5} - 1)}{6}}.$$

21. The spherical band has $\pi/4 \le \phi \le 3\pi/4$, so

$$A = \int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \sin\phi\ \mathrm{d}\phi\ \mathrm{d}\theta = \boxed{2\sqrt{2}\,\pi}.$$

22. $\boxed{3\sqrt{14}}$.

23. With $\mathbf{r}(\phi,\theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi)$, the outward vector area element is $\sin\phi\,\mathbf{r}\,\mathrm{d}\phi\,\mathrm{d}\theta$. The complete setup is

$$\boxed{\int_0^{2\pi}\int_0^{\pi}\left(\sin^3\phi\cos^2\theta\sin\theta + 2\sin\phi\sin\theta - \cos^2\phi\right)\sin\phi\;\mathrm{d}\phi\;\mathrm{d}\theta}.$$

24. The downward vector area element is $(r\cos\theta, r\sin\theta, -r)\;\mathrm{d}r\;\mathrm{d}\theta$, giving

$$\int_0^{2\pi}\int_2^4 (r^2\sin\theta\cos\theta - r)\;\mathrm{d}r\;\mathrm{d}\theta = \boxed{-12\pi}.$$

25. The upward graph normal factor is $(1/3, 1, 1)$, and $(3, 1, -2)\cdot(1/3, 1, 1) = 0$. Thus the flux is $\boxed{0}$.

26. The plane is $z = 6 - 3x/5 - 2y/5$, and the upward flux integrand is $(3y + 7x)/5$ over the triangle with vertices $(0,0), (10,0), (0,15)$:

$$\int_0^{10}\int_0^{15-3x/2}\frac{3y+7x}{5}\;\mathrm{d}y\;\mathrm{d}x = \boxed{575}.$$

27. With $z_x = 2xy^2 + 1$ and $z_y = 2x^2y + 1$, the setup is

$$\boxed{\int_0^1\int_0^1 (-xy^2 - 4x^2y - 3)\;\mathrm{d}x\;\mathrm{d}y}.$$

Its value, for a check, is $-23/6$.

28. With $z_x = 4x^3$ and $z_y = 2y$, the setup is

$$\boxed{\int_{-1}^1\int_{-1}^1 \left(-4x^5 - 2y(x^4+y^2) + (x^4+y^2)^2\right)\;\mathrm{d}x\;\mathrm{d}y}.$$

Its value, for a check, is $16/9$.

29. $\boxed{-\pi}$. The intended surface includes its equatorial rim ($z \geq 0$); the rim is the boundary used in Stokes' theorem.

30. $\boxed{0}$.

31. $\boxed{0}$.

32. $\boxed{-5\pi}$.

33. $\boxed{0}$.

34. $\boxed{0}$. The hemisphere together with its base disk is closed, so its boundary is empty.

35. $\boxed{0}$. The entire cube surface is closed.

36. Valid parametrizations and bounds are:
    (a) $\mathbf{r}(u,v) = (u, v, 10)$, $-1 \le u \le 1$, $-2 \le v \le 2$. The stated bounds describe a rectangle, rather than a square.
    (b) $\mathbf{r}(u,v) = (u, v, u^2 + v^2)$, $0 \le v \le 1$, $0 \le u \le v$.
    (c) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, r^2)$, $0 \le r \le 3$, $0 \le \theta \le 2\pi$.
    (d) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, r)$, $2 \le r \le 4$, $0 \le \theta \le 2\pi$.
    (e) $\mathbf{r}(r,\theta) = (r\cos\theta, r\sin\theta, 4 - r\cos\theta + r\sin\theta)$, $0 \le r \le 3$, $0 \le \theta \le 2\pi$.
    (f) $\mathbf{r}(\theta,z) = (\cos\theta, \sin\theta, z)$, $0 \le \theta \le \pi/2$, $0 \le z \le 3$.
    (g) $\mathbf{r}(\theta,z) = (2\cos\theta, 2\sin\theta, z)$, $0 \le \theta \le 2\pi$, $2\sin\theta - 2 \le z \le 2\cos\theta + 4$.
    (h) $\mathbf{r}(\phi,\theta) = (2\sin\phi\cos\theta, 2\sin\phi\sin\theta, 2\cos\phi)$, $0 \le \phi \le \pi/4$, $0 \le \theta \le 2\pi$.

37. $\boxed{\text{(a), (d), (e)}}$ under the course convention that ∇ denotes the gradient of a *scalar* field; a Jacobian of a vector field is not being requested in option (c).

38. $\boxed{\text{(a), (b), (d)}}$.

39. $\boxed{\text{(a), (b), (d)}}$.

40. In order: $\boxed{\text{F, T, T, F, F, T}}$.

41. Choices are (a) direct parametrization, (b) FTLI, (c) Green, (d) Stokes, (e) none. The coordinator's intended *direct-method* selections are listed below. The original question does not explicitly exclude planar reductions; those also make Green valid in (3) and (5).

Case	Direct selections	Allowing planar Green	Integral value
(1)	(a), (c), (d)	same	2π
(2)	(a), (b), (c), (d)	same	0
(3)	(a), (d)	(a), (c), (d)	0
(4)	(a), (b)	same	2
(5)	(a), (d)	(a), (c), (d)	$-3/2$
(6)	(a), (b), (c), (d)	same	0

In (2), (4), and (6), potentials are x^2y, x^2y+xz, and x^3y+y^2, respectively. In (4) the curve is open. In (1) the planar scalar curl is 2. The three-dimensional curls in (3) and (5) are $(1,1,1)$ and $(-1,-1,-1)$, respectively, so neither field is conservative.

For (3), substitute $z = y + 2$, $\mathrm{d}z = \mathrm{d}y$ into the line integral:

$$\mathbf{F}\cdot\mathrm{d}\mathbf{r} = (y+2)\,\mathrm{d}x + (x+y)\,\mathrm{d}y, \qquad Q_x - P_y = 0.$$

For (5), substitute $z = 1 - x - y$, $\mathrm{d}z = -\,\mathrm{d}x - \mathrm{d}y$:

$$\mathbf{F}\cdot\mathrm{d}\mathbf{r} = (y-x)\,\mathrm{d}x + (1-2x-y)\,\mathrm{d}y, \qquad Q_x - P_y = -3.$$

Thus both curves are planar (in tilted planes), and Green can be applied after these explicit reductions. The base triangle in (5) has area 1/2.

Exam3 Practice Problems

1. Find the curl and divergence. Use curl to determine whether the vector field is conservative.
   (a) $\mathbf{F} = x^2y\,\mathbf{i} - (z^3 - 3x)\,\mathbf{j} + 4y^2\,\mathbf{k}$
   (b) $\mathbf{F} = (3x + 2z^2)\,\mathbf{i} + (\frac{x^3y^2}{z})\,\mathbf{j} - (z - 7x)\,\mathbf{k}$
   (c) $\mathbf{F} = 6x\,\mathbf{i} + (2y - y^2)\,\mathbf{j} + (6z - x^3)\,\mathbf{k}$
   (d) $\mathbf{F} = (z\cos y)\,\mathbf{i} - (xz\sin y)\,\mathbf{j} + (x\cos y)\,\mathbf{k}$
2. Determine whether or not the vector $\mathbf{F}(x, y, z) = (2xy^2 + z)\mathbf{i} + (2yx^2)\mathbf{j} + x\mathbf{k}$ is conservative. If it is, compute the potential function.
3. Determine whether or not the vector $\mathbf{F}(x, y, z) = (yz + e^x)\mathbf{i} + (xz + e^y)\mathbf{j} + (xy + e^z)\mathbf{k}$ is conservative. If it is, compute the potential function.
4. Evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$ and $\mathbf{F} = (yz^2)\mathbf{i}+(xz^2)\mathbf{j}+(2xyz)\mathbf{k}$, where C is a path from $(0, 0, 0)$ to $(1, 1, 1)$.
5. Evaluate $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$ and $\mathbf{F} = (yz + e^x)\mathbf{i} + (xz + e^y)\mathbf{j} + (xy + e^z)\mathbf{k}$, where C is a path from $(0, 0, 0)$ to $(1, 1, 1)$.
6. Let $\mathbf{F} = y^2\mathbf{i} - x\mathbf{j}$, and let C be a triangle with vertices $(0, 0), (1, 0), (1, 1)$ traversed counterclockwise. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
7. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
8. Let $\mathbf{F} = (y + x)\mathbf{i} + (y - x)\mathbf{j}$, and let C be a circle with $x^2 + y^2 = 4$ traversed counterclockwise. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
9. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
10. Let $\mathbf{F} = (5x)\mathbf{i} + (x + 3y)\mathbf{j}$, and let C be the square obtained by going from $(0, 0)$ to $(1, 0)$, from $(1, 0)$ to $(1, 1)$, from $(1, 1)$ to $(0, 1)$, and from $(0, 1)$ to $(0, 0)$ (traveling along a straight horizontal or vertical line each time). Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
11. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
12. Let $\mathbf{F} = (x^2)\mathbf{i} + (yx)\mathbf{j}$, and let C be curve obtained by going from $(0, 0)$ to $(1, 1)$ along the graph of the curve $y = x^2$, and then from $(1, 1)$ to $(0, 0)$ along the graph of $y = x$. Use Green's theorem to evaluate the integral $\int_C \mathbf{F}\cdot\mathbf{T}\ ds$
13. Repeat the previous problem, but now compute $\int_C \mathbf{F}\cdot\mathbf{N}\ ds$
14. Consider the surface $\mathbf{r}(u, v) = \langle u + v, u^2 - v^2, uv\rangle$, for $0 \le u \le 1$, $0 \le v \le 2$. Set up, but do not solve, an integral which represents the area of this surface.
15. Consider the surface $\mathbf{r}(u, v) = \langle u^2v, uv^2, u + v\rangle$, for $0 \le u \le 1$, $0 \le v \le 1$. Suppose the density of this surface is given by the function $\delta(u, v) = e^{u+v}$. Set up, but do not solve, an integral which represents the mass of this surface.

16. Consider the surface obtained by revolving the graph of the function $y = e^{5x}$, for $0 \leq x \leq 1$, around the x-axis. Set up, but do not solve, an integral which represents the area of this surface.

17. Consider the surface obtained by revolving the graph of the function $y = 1/\sqrt{x}$, for $1 \leq x \leq 4$, around the y-axis. Let $f(x, y) = y^5$. Set up, but do not evaluate, the surface integral $\int\int_S f \, d\sigma$.

18. Consider the surface consisting of the portion of the cylinder $x^2 + y^2 = 1$ which is above $z = 0$ and below $z = 1$. Let $f(x, y, z) = x^2 z^2$. Evaluate the surface integral $\int\int_S f \, d\sigma$.

19. Consider the surface consisting of the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 2$ and below the plane $z = 4$. Let $f(x, y, z) = z^2$. Evaluate the surface integral $\int\int_S f \, d\sigma$

20. Consider the surface consisting of the portion of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 1$. Find the area of this surface.

21. Consider the surface consisting of the portion of the sphere $x^2 + y^2 + z^2 = 1$ which is below the upward facing cone $z = \sqrt{x^2 + y^2}$ and above the downward facing cone $z = -\sqrt{x^2 + y^2}$. Find the area of this surface.

22. Consider the surface consisting of the portion of the plane $2x + 3y + z = 6$ which is in the first octant. Find the area of this surface.

23. Let S be the outward oriented sphere $x^2 + y^2 + z^2 = 1$, and let $\mathbf{F} = xy\mathbf{i} + 2\mathbf{j} - z\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

24. Let S be the surface consisting of the portion of the cone $z = \sqrt{x^2 + y^2}$ which is above the plane $z = 2$ and below the plane $z = 4$, oriented so that the normal is always pointing down. Let $\mathbf{F} = y\mathbf{i} + 1\mathbf{k}$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, dS$.

25. Let S be the surface given by the portion of the plane $x + 3y + 3z = 9$ which lies in the first octant, oriented so that the normal always points in the positive z direction. Let $\mathbf{F} = \langle 3, 1, -2 \rangle$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

26. Let S be the surface consisting of the portion of the plane $3x + 2y + 5z = 30$ which lies in the first octant, oriented upwards. Let $\mathbf{F} = \langle y, x, x \rangle$ be a vector field. Compute the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

27. Consider the surface consisting of the portion of the graph of $z = x^2 y^2 + x + y$ which is above the square $0 \leq x \leq 1$, $0 \leq y \leq 1$, oriented up. Let $\mathbf{F}(x, y) = 1\mathbf{i} + 2\mathbf{j} + xy^2\mathbf{k}$. Set up, but do not evaluate, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

28. Let S be the surface given by the portion of the graph of $z = x^4 + y^2$ which is above the square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, oriented up. Let $\mathbf{F} = x^2\mathbf{i} + z\mathbf{j} + z^2\mathbf{k}$ be a vector field. Set up, but do NOT solve, the flux integral $\int\int_S \mathbf{F} \cdot \mathbf{N} \, d\sigma$.

29. Let S be the outward oriented top half of the sphere $x^2 + y^2 + z^2 = 1$ for $z > 0$. Let $\mathbf{F} = \langle y, 0, -z^2 \rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$.

30. Let S be the downward oriented surface consisting of the portion of the sphere $x^2 + y^2 + z^2 = 4$ which lies below the plane $z = -1$. Let $\mathbf{F} = \langle 1 - z, -3y, x \rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S (\nabla \times \mathbf{F}) \cdot \mathbf{N} \, d\sigma$.

31. Let S be the outward oriented surface given by the portion of the cylinder $x^2+y^2=1$ which is above the plane $z=-1$ and below the plane $z=1$. Let $\mathbf{F}=\langle z, z^2, z^3\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

32. Let S be the inward surface given by the portion of the cylinder $x^2+y^2=1$ which is above the plane $z=0$ and below the plane $z=5$. Let $\mathbf{F}=\langle yz, 0, 2z\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

33. Let S be the downward oriented surface consisting of the portion of the cone $z=\sqrt{x^2+y^2}$ which is above the plane $z=1$ and below $z=3$. Let $\mathbf{F}=\langle 1-z, -3y, x\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

34. Let S be the outward oriented surface consisting of the top hemisphere $x^2+y^2+z^2=25$ and the disk area $x^2+y^2\leq 25$ at height $z=0$. Let $\mathbf{F}=\langle x^2y, x^3, yx^2\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

35. Let S be the outward oriented cube with vertices $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, $(1,1,0)$, $(1,0,1)$, $(0,1,1)$ and $(1,1,1)$. Let $\mathbf{F}=\langle y^2, xy^2, xz^2\rangle$ be a vector field. Use Stokes' theorem to compute $\int\int_S(\nabla\times\mathbf{F})\cdot\mathbf{N}\,d\sigma$.

36. Find a parameterization for each surface:

    (a) The portion of the surface $z=10$ that is above the square $-1\leq x\leq 1$, $-2\leq y\leq 2$.

    (b) The paraboloid $z=x^2+y^2$ that is above the triangle with vertices at (0,0), (0,1) and (1,1).

    (c) The paraboloid $z=x^2+y^2$ that is above the circular disk of radius 3 centered at the origin.

    (d) The portion of the cone $z=\sqrt{x^2+y^2}$ that lies between the cylinders $x^2+y^2=4$, $x^2+y^2=16$.

    (e) The portion of the plane $x-y+z=4$ that is within the cylinder $x^2+y^2=9$.

    (f) The portion of cylinder $x^2+y^2=1$ that lies between the planes $z=0$ and $z=3$ in the first octant.

    (g) The portion of cylinder $x^2+y^2=4$ that lies between the plane $z=y-2$ and $z=x+4$.

    (h) The portion of the sphere $x^2+y^2+z^2=4$ that lies inside the cylinder $x^2+y^2=2$ and above the xy-plane.

37. Let f be a scalar function defined on $\mathbb{R}^3$ and let $\mathbf{F}$ be a vector field defined on $\mathbb{R}^3$. Assume that f and all components of vector field $\mathbf{F}$ have continuous second partial derivatives. Select all expressions that are well-defined.

    (a) $\nabla\cdot(\nabla\times\mathbf{F})$

    (b) $\nabla\times(\nabla\cdot\mathbf{F})$

    (c) $\nabla(\nabla\times\mathbf{F})$

    (d) $\nabla\times(\nabla\times\mathbf{F})$

    (e) $(\nabla f)\cdot(\nabla\times\mathbf{F})$

    (f) $\nabla\cdot(\nabla\cdot\mathbf{F})$

    (g) None of the above are correct

38. Select all true statements about curl.

    (a) $\operatorname{curl}\mathbf{F}$ at a point measures the tendency of the field to rotate around that point.
    (b) The $\operatorname{curl}\mathbf{F}$ points along the axis of rotation.
    (c) If $\operatorname{curl}\mathbf{F} = 0$ everywhere, the field $\mathbf{F}$ is constant.
    (d) For $\mathbf{F} = \langle -y, x, 0\rangle$, $\operatorname{curl}\mathbf{F} = 2\mathbf{k}$.

39. Select all true statements about divergence.

    (a) $\operatorname{div}\mathbf{F} > 0$ at a point means that point acts like a source.
    (b) If $\operatorname{div}\mathbf{F} = 0$ everywhere, the field is incompressible.
    (c) If $\operatorname{div}\mathbf{F} = 0$ everywhere, then $\mathbf{F}$ must also have $\operatorname{curl}\mathbf{F} = \mathbf{0}$.
    (d) For $\mathbf{F} = \langle x, y, z\rangle$, $\operatorname{div}\mathbf{F} = 3$ at every point.

40. Determine whether the following statements are true or false.

    (a) The fundamental theorem of line integrals can only be applied to closed curves.
    (b) The fundamental theorem of line integrals can only be applied to conservative vector fields.
    (c) Let C be a simple closed curve in the xy-plane, oriented counterclockwise, that encloses a region R, and let $\mathbf{F} = <P, Q>$ where P and Q have continuous partial derivatives on R. Then $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$ can be evaluated using Green's Theorem, whether $\mathbf{F}$ is conservative or not.
    (d) If C is a closed curve, then $\int_C \mathbf{F}\cdot\mathbf{T}\ \ ds = 0$ for any vector field $\mathbf{F}$.
    (e) If $\mathbf{F}$ is a vector field, then $div\,\mathbf{F}$ is a vector field.
    (f) If $\mathbf{F}$ is a vector field, then $curl\,\mathbf{F}$ is a vector field.

41. Consider the following methods

    (a) Direct parameterization
    (b) Fundamental Theorem of Line Integrals
    (c) Green's Theorem
    (d) Stokes' Theorem
    (e) None of the methods without more information

    Select all methods that could be used to evaluate

    (1) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle x^2 - y,\ x + y^2\rangle$ and C is the unit circle in the xy-plane (counterclockwise).
    (2) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 2xy,\ x^2\rangle$ and C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$, traversed counterclockwise.
    (3) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle z,\ x,\ y\rangle$ and C is the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 2$, traversed counterclockwise when viewed from above.
    (4) $\int_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 2xy + z,\ x^2,\ x\rangle$ and C is the curve $\mathbf{r}(t) = \langle t, t^2, t^3\rangle$ for $0 \le t \le 1$ (from $(0,0,0)$ to $(1,1,1)$).
    (5) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle y,\ z,\ x\rangle$ and C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, traversed counterclockwise when viewed from above.
    (6) $\oint_C \mathbf{F}\cdot\mathbf{T}\,ds$, where $\mathbf{F} = \langle 3x^2y,\ x^3 + 2y\rangle$ and C is the circle $x^2 + y^2 = 4$ in the xy-plane, traversed counterclockwise.